

Answer Key — Biology Grade 11

Sub-Strand 3.1: Reproduction in Animals • Lesson 3: Built for the Journey: Structure of Sperm and Ovum

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7
D	A	D	B	C	C	A

USE AFTER: PREDICT PHASE — Predict: sketching a sperm and an ovum and predicting which is bigger

Q1. Before looking at any slides, a learner predicts that the human ovum will be much larger than a sperm cell. Which reason best supports this prediction?

- A. The ovum must be big enough to be seen by the sperm
- B. The ovum has to swim a long distance, so it needs a big tail
- C. The ovum has twice as many chromosomes as the sperm, so it takes up more space
- D. The ovum stores food in its cytoplasm and has a protective layer, but it does not swim**

Correct answer: D. The ovum does not move. It is large because it carries food reserves in its cytoplasm and has a protective layer (jelly coat/zona pellucida) to support the early embryo.

USE AFTER: OBSERVE PHASE — Observe: drawing and labelling the sperm (midpiece with mitochondria)

Q2. Under a microscope, a learner sees that the midpiece of a sperm cell is packed with mitochondria. What is the function of this structure?

- A. It supplies the energy that the flagellum needs for swimming**
- B. It releases enzymes that digest the outer layer of the ovum
- C. It protects the sperm from being damaged inside the female body
- D. It stores the chromosomes that will be passed to the offspring

Correct answer: A. Mitochondria release energy by respiration. In the midpiece this energy powers the beating of the flagellum (tail) so the sperm can swim towards the ovum.

USE AFTER: OBSERVE PHASE — Observe: observation table comparing size of sperm and ovum

Q3. A human sperm cell is about 50 micrometres long. A human ovum is about 0.1 mm across (1 mm = 1000 micrometres). About how many times wider is the ovum than the sperm is long?

- A. 5 times
- B. 200 times
- C. 20 times
- D. 2 times**

Correct answer: D. Convert 0.1 mm to micrometres: $0.1 \times 1000 = 100$ micrometres. Then $100 \div 50 = 2$, so the ovum is about 2 times the size of the sperm's length.

USE AFTER: EXPLAIN PHASE — Explain: CER frame on the functions of the acrosome, flagellum, mitochondria and jelly coat

Q4. During fertilisation, the acrosome at the head of the sperm releases enzymes. What do these enzymes do?

- A. They give the sperm energy to swim
- B. They digest the jelly coat (zona pellucida) around the ovum so the sperm can enter**
- C. They make the ovum produce more food reserves
- D. They join the sperm nucleus to the ovum nucleus

Correct answer: B. The acrosome contains digestive enzymes that break down the jelly coat/zona pellucida, allowing the sperm head to reach and fuse with the ovum. Energy for swimming comes from the mitochondria.

USE AFTER: EXPLAIN PHASE — Explain: CER frame on the haploid nucleus of each gamete

Q5. Both the sperm and the ovum have haploid nuclei (half the normal number of chromosomes). Why is this important for fertilisation?

- A. Haploid nuclei let the ovum divide without a sperm
- B. Haploid nuclei make the sperm lighter so it can swim faster
- C. The fused nuclei give the zygote the full normal number of chromosomes**
- D. Haploid nuclei mean the offspring will have only half the genes of the mother

Correct answer: C. When the haploid sperm nucleus fuses with the haploid ovum nucleus, the zygote gets the full normal chromosome number. Without haploid gametes, the number would double in every generation.

USE AFTER: EXPLAIN PHASE — Explain: Think-Pair-Share on why frogs need thousands of eggs but cows only one calf

Q6. A frog at the Lake Victoria wetlands releases thousands of eggs, but a cow usually gives birth to one calf. Which explanation best accounts for this difference?

- A. Cows cannot make more than one ovum because they have no jelly coat
- B. Frogs have more sperm than cows, so they must also make more eggs
- C. Frog eggs are fertilised in open water and have little protection, so many are lost, while a cow's fertilised egg is protected and nourished inside her body**
- D. Frog eggs are bigger than cow eggs, so more of them are needed

Correct answer: C. External fertilisation in water exposes frog eggs to predators and other dangers, so many eggs are needed for a few to survive. Internal fertilisation and development inside the cow protects the embryo, so one calf is enough.

USE AFTER: MODEL BUILDING PHASE — Model: adding the arrow 'IVF: fertilisation in dish, embryo then placed in womb' to the class model

Q7. A news report says that in IVF the baby grows in a test tube from start to birth. Which statement correctly describes IVF?

- A. Fertilisation happens in a laboratory dish, and the embryo is then placed in the womb of the mother or a surrogate to develop**
- B. Fertilisation and the whole development of the baby happen in the laboratory
- C. The ovum develops into a baby in the dish without any sperm
- D. Fertilisation happens in the womb, and only the sperm is prepared in the laboratory

Correct answer: A. In IVF only fertilisation happens outside the body, in a dish. The embryo is then placed in the womb of the mother or a surrogate, where it grows until birth.